WASTE PARTICLE REINFORCED EPOXY COMPOSITE

A PROJECT REPORT

Submitted by

DNEENADHAYALAN S 8115U24ME010

in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

In

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

CHENNAI 600 025

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MEB1225- UG PROJECT WORK

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Under the Guidance of

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MECHANICAL ENGINEERING

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DECLARATION BY THE CANDIDATE

I declare that to the best of our knowledge the work reported here in has been composed solely by ourself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

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ACKNOWLEDGEMENT

I thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

At this pleasing moment of having successfully completed our project, we wish to convey our sincere thanks and gratitude to our beloved kind Chairman, **Dr. K. RAMAKRISHNAN, B.E., Ph.D.,** who provided all the facilities to us.

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PO11 Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context

ABSTRACT

This project presents the development and tensile strength analysis of a waste particle reinforced epoxy composite using used tea powder and ramie fibre as reinforcement materials. The composite specimen is fabricated using the hand lay-up method with epoxy resin and hardener as the matrix. The main objective of this study is to evaluate the mechanical strength of the developed eco-friendly composite material. A tensile test is conducted using a Universal Testing Machine (UTM) to determine key properties such as tensile strength, elongation, and load-bearing capacity. The results obtained from the test are analyzed to understand the behavior of the composite under tensile loading. The study highlights the effective utilization of waste tea powder and natural fibre in improving material properties while promoting sustainability. The developed composite shows potential for lightweight and low-load engineering applications. Overall, this project contributes to the development of cost-effective and environmentally friendly composite materials.

KEYWORDS: ***Epoxy Resin, Waste Tea Powder, Ramie Fibre, Composite Material, Tensile Strength, Hand Lay-Up Technique, Natural Fibre Reinforcement***

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CHAPTER 1

INTRODUCTION

1.1 Organic composite material:

Epoxy resin is a widely used thermosetting polymer that acts as an effective matrix material in composite manufacturing due to its strong adhesion, high strength, and good chemical resistance. It is formed by the reaction between epoxy groups and a hardener, resulting in a rigid, cross-linked structure after curing. Once set, epoxy cannot be remelted, which makes it stable and suitable for structural applications. In composites, epoxy resin binds the reinforcement materials, transfers applied loads, and protects them from environmental effects.

One of the main advantages of epoxy resin is its excellent bonding capability with materials such as natural fibres and particulate fillers like used tea powder. This ensures proper load transfer between the matrix and reinforcement, improving the overall mechanical properties of the composite. Epoxy also has low shrinkage during curing, which helps maintain dimensional accuracy and reduces internal stresses. Additionally, it offers good resistance to moisture and chemicals.

1.2 Composite Material Composition:

Epoxy resin is a widely used thermosetting polymer that acts as an effective matrix material in composite manufacturing due to its strong adhesion, high strength, and good chemical resistance. It is formed by the reaction between epoxy groups and a hardener, resulting in a rigid, cross-linked structure after curing. Once set, epoxy cannot be remelted, which makes it stable and suitable for structural applications. In composites, epoxy resin binds the reinforcement materials, transfers applied loads, and protects them from environmental effects.

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The curing process involves mixing epoxy resin with a hardener in a specific ratio, which initiates a chemical reaction leading to solidification. The final properties depend on proper mixing, curing time, and temperature. Incorrect proportions or poor mixing can result in weak bonding and reduced strength. Although epoxy resin is strong, it is relatively brittle, so it is often combined with reinforcement materials to improve toughness.

In this project, epoxy resin is used as the matrix material to develop a composite reinforced with used tea powder and ramie fibre. It plays a key role in binding the materials, distributing stress, and enhancing the strength and durability of the composite. Overall, epoxy resin is essential in producing lightweight, strong, and eco-friendly composite materials.



Fig 1.1 Epoxy Resin

1.3 Composite Reinforcement:

Reinforcement in composite materials involves improving the mechanical properties of the base material by adding strengthening elements. In this project, reinforcement is achieved using used tea powder and ramie fibre in an epoxy matrix. Unlike synthetic reinforcements such as carbon or glass fibres, these natural and waste materials provide an eco-friendly and cost-effective alternative. Tea powder acts as a particulate filler that improves stiffness and reduces material cost, while ramie fibre acts as a primary reinforcement, significantly enhancing tensile strength and load-bearing capacity due to its fibrous structure.

The addition of ramie fibre improves the composite's ability to resist deformation and increases its durability under tensile loading. Tea powder, being fine and porous, helps in filling voids within the matrix and enhances the overall density and bonding characteristics of the composite. Together, these reinforcements contribute to better stress distribution and improved interfacial bonding with the epoxy resin.

Although epoxy resin itself has good adhesive properties, it is relatively brittle. By incorporating tea powder and ramie fibre, the composite gains improved toughness, stiffness, and resistance to mechanical failure. The effectiveness of reinforcement depends on factors such as proper mixing, uniform distribution, fibre orientation, and good bonding between matrix and reinforcement. Poor dispersion or excess moisture can lead to weak bonding and reduced strength.

In this project, the use of natural and waste-based reinforcements not only enhances mechanical performance but also promotes sustainability. The developed composite demonstrates that materials like tea powder and ramie fibre can effectively replace conventional reinforcements for low to moderate load applications. Thus, reinforcement plays a crucial role in improving the overall performance, reliability, and applicability of the developed epoxy composite material.

1.4 Factors Contributing to Mechanical Properties:

The mechanical properties of the developed waste particle reinforced epoxy composite are influenced by several important factors related to material composition and fabrication. The resin-to-hardener ratio plays a crucial role, as the correct proportion ensures complete curing and strong bonding, while an improper ratio can lead to weak and brittle material. The amount of tea powder added affects stiffness and cost; a moderate quantity improves performance, but excessive addition may reduce bonding strength and increase brittleness. Similarly, the content and orientation of ramie fibre significantly influence tensile strength, where higher fibre content and proper alignment enhance load-bearing capacity.

The quality of mixing is also important, as uniform distribution of tea powder and fibres ensures good interfacial bonding and minimizes defects such as voids or weak zones. In addition, curing time and temperature must be properly controlled to achieve complete cross-linking of the epoxy resin, which directly affects strength and durability. Factors like moisture content in tea powder, fibre dispersion, and fabrication technique (hand lay-up) also impact the final properties of the composite. Overall, careful control of these factors is essential to obtain a composite material with improved mechanical performance and reliability.

1.5 Properties of Composite Material:

Mechanical Properties

The epoxy composite material exhibits good tensile strength suitable for low to moderate load applications. It has high stiffness due to the rigid epoxy matrix combined with the reinforcing effect of ramie fibre. Although epoxy is naturally brittle, the addition of ramie fibre improves toughness and reduces sudden failure. The composite shows low to moderate elongation before breaking, depending on fibre content and distribution. Its strength mainly depends on the quality of bonding between

epoxy, tea powder, and fibre, as well as uniform mixing and proper curing conditions.

Thermal Properties

The epoxy composite has moderate thermal stability and does not melt like thermoplastics, as it is a thermosetting material. Instead, it undergoes degradation at high temperatures after curing. During fabrication, heat is generated in the curing process, forming a strong cross-linked structure. The material can withstand normal and slightly elevated temperatures but may soften or degrade when exposed to excessive heat. Therefore, it is mainly suitable for room temperature and low to moderate thermal applications.

Physical Properties

The developed composite is lightweight and easy to handle, making it suitable for practical applications. It has good rigidity and maintains its shape after curing due to the strong epoxy matrix. The surface finish depends on the mould and fabrication method, with hand lay-up typically producing a slightly rough texture. The presence of tea powder gives a natural appearance, while ramie fibre improves the internal structure and strength. Overall, the composite is eco-friendly as it utilizes waste tea powder and natural fibre.

Chemical Properties

The epoxy composite shows good resistance to water, oils, and mild chemicals under normal conditions. It is more stable compared to natural fibres alone, as the epoxy matrix protects the reinforcement materials. However, prolonged exposure to moisture can affect the bonding between fibre and matrix. The material has moderate resistance to acids and alkalis but is not suitable for highly corrosive environments. Under normal atmospheric conditions, the composite remains stable and durable.

Electrical Properties

The developed waste particle reinforced epoxy composite is an eco-friendly material known for its good strength, strong bonding, and suitability for lightweight engineering applications. The combination of epoxy resin with used tea powder and ramie fibre improves stiffness and mechanical performance while promoting sustainability. The composite exhibits moderate tensile strength and high rigidity, making it suitable for low to moderate load applications. However, due to the brittle nature of epoxy, the material may show limited impact resistance, although the addition of ramie fibre helps in improving toughness and reducing sudden failure.

Unlike thermoplastics, this composite does not melt, as epoxy is a thermosetting material, but it can degrade at high temperatures. Therefore, it is mainly suitable for room temperature and moderate thermal conditions. The material provides good dimensional stability after curing, though the surface finish depends on the fabrication method used. It shows resistance to water, oils, and mild chemicals, but prolonged exposure to moisture may affect fibre-matrix bonding. Overall, the developed composite is a cost-effective and environmentally friendly material suitable for experimental and light engineering applications, and its properties can be improved by optimizing composition, mixing, and curing conditions.

1.6 Advantages and Disadvantages of Epoxy Composite Material :

Advantages

- Eco-friendly material as it utilizes waste tea powder and natural ramie fibre.
- Good tensile strength and stiffness due to fibre reinforcement.
- Strong bonding properties of epoxy ensure good load transfer between materials.
- Lightweight and suitable for low to moderate load applications.
- Cost-effective due to the use of waste and natural materials.
- Good resistance to water, oils, and mild chemicals.

Disadvantages

- Epoxy matrix is brittle and may fail suddenly under high impact loads.
- Not suitable for high-temperature applications as it can degrade under excessive heat.
- Moisture absorption by natural fibres may affect bonding and strength over time.
- Requires proper mixing and curing; otherwise, defects like voids and weak bonding may occur.
- Mechanical properties depend on uniform distribution of tea powder and fibre.
- Not suitable for heavy-load or highly corrosive environments.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction:

Composite materials have gained significant attention in recent years due to their ability to combine different materials to achieve improved mechanical properties while maintaining low weight and cost. Among various composite systems, polymer matrix composites using epoxy resin are widely preferred because of their strong bonding characteristics, good strength, and resistance to environmental effects. In recent times, there has been increasing interest in developing eco-friendly composites by incorporating natural fibres and waste materials as reinforcement, reducing environmental impact and material cost.

Used tea powder, an agricultural waste, and ramie fibre, a natural fibre with good tensile strength, have shown potential as sustainable reinforcement materials in polymer composites. The mechanical properties of such composites depend on several factors, including the proportion of reinforcement, mixing quality, fibre distribution, and curing conditions. Proper optimization of these factors can significantly improve tensile strength, stiffness, and durability of the developed material.

Several studies have focused on natural fibre reinforced composites, but limited work has been done on utilizing waste materials like tea powder in combination with natural fibres for structural applications. Therefore, this project aims to develop a waste particle reinforced epoxy composite using tea powder and ramie fibre and to evaluate its tensile strength using a Universal Testing Machine (UTM). This study helps in understanding the behavior of eco-friendly composite materials under tensile loading and highlights their potential for lightweight and low-cost engineering applications.

2.2 Inferences of the Literature:

- Proper resin-to-hardener ratio is essential for complete curing; correct proportions improve bonding and tensile strength, while improper ratios lead to weak and brittle composites.
- The percentage of tea powder significantly affects properties; moderate addition improves stiffness and reduces cost, but excessive content can reduce bonding and create weak zones.
- Ramie fibre plays a major role in strength enhancement; higher fibre content and proper alignment improve tensile strength and load-bearing capacity.
- Uniform mixing and proper dispersion of tea powder and fibres are critical, as poor distribution can cause voids, weak interfacial bonding, and reduced mechanical performance.
- Curing conditions such as time and temperature greatly influence the final properties; adequate curing ensures strong cross-linking, while insufficient curing results in lower strength.
- Natural fibre composites show good stiffness and eco-friendly benefits but may face issues like moisture absorption and variability in properties.
- Most research studies focus on synthetic fibre composites or standard specimens, with limited work on waste-based reinforcements like tea powder.

2.3 Problem Statement:

Although epoxy-based composites are widely used in engineering applications due to their strong bonding characteristics, good mechanical strength, and chemical resistance, the development of eco-friendly composites using waste materials is still an emerging area. In recent years, there has been increasing interest in utilizing agricultural waste and natural fibres to produce sustainable composite materials. However, the mechanical performance of such waste particle reinforced epoxy composites is not fully understood and requires detailed experimental investigation. In particular, the combination of used tea powder as a particulate filler and ramie fibre as a natural reinforcement introduces complexity in material behavior, especially in terms of bonding, load transfer, and durability.

The strength and performance of the developed composite are highly dependent on several factors such as resin-to-hardener ratio, percentage of tea powder, fibre content, mixing quality, and curing conditions. Improper selection or control of these parameters can lead to defects such as poor interfacial bonding, formation of air voids, uneven distribution of reinforcement, and incomplete curing, all of which significantly reduce the mechanical properties of the composite. Additionally, excessive addition of tea powder may weaken the matrix by reducing effective bonding, while insufficient fibre content may limit the reinforcement effect, leading to lower tensile strength.

Natural fibres like ramie also present certain challenges, including moisture absorption and variability in properties, which can affect the long-term durability and consistency of the composite material. Furthermore, most existing research studies are focused on synthetic fibre reinforced composites such as glass or carbon fibre systems, with relatively limited work carried out on hybrid composites incorporating waste materials like tea powder. There is also a lack of standardized data and clearly defined optimal composition for achieving maximum strength in such eco-friendly composites.

Another important limitation is the scarcity of experimental data on the tensile

behavior of these materials under loading conditions. Without proper testing and analysis, it is difficult to evaluate their suitability for practical engineering applications. Therefore, there is a need to systematically develop and test waste particle reinforced epoxy composites to understand their mechanical behavior and identify optimal material composition. This project aims to address these gaps by fabricating a composite using epoxy resin, used tea powder, and ramie fibre, and evaluating its tensile strength using a Universal Testing Machine (UTM), thereby contributing to the development of sustainable and cost-effective composite materials.

* he mechanical strength of the developed epoxy composite is highly dependent on factors such as resin-to-hardener ratio, tea powder percentage, fibre content, and mixing quality, which can lead to inconsistency in performance if not properly controlled.

* Improper mixing and uneven distribution of tea powder and ramie fibre can result in poor interfacial bonding, formation of air voids, and weak zones, ultimately reducing tensile strength and durability.

* Excess addition of tea powder may act as a defect instead of reinforcement, decreasing bonding efficiency and increasing brittleness of the composite material.

* Curing conditions such as insufficient curing time or incorrect resin-to-hardener ratio can lead to incomplete cross-linking, resulting in poor mechanical properties and reduced strength.

* Most existing research focuses on synthetic fibre composites (like glass or carbon fibre), while limited studies are available on hybrid composites using waste materials like tea powder combined with natural fibres.

* There is no clearly established optimal composition (resin, filler, and fibre ratio) for achieving maximum strength and performance in such eco-friendly .

CHAPTER – 3

MATERIALS AND METHODOLOGY

3.1 Experimental Methodology:

The materials selected for the development of the composite include epoxy resin as the primary matrix material, hardener as the curing agent, used tea powder as a waste particulate filler, and ramie fibre as the natural reinforcement. Epoxy resin is chosen due to its excellent bonding properties, mechanical strength, and chemical resistance. The hardener is used to initiate the curing reaction, which converts the liquid resin into a solid cross-linked structure. Used tea powder is utilized as a low-cost, eco-friendly filler material that enhances stiffness and reduces material cost, while ramie fibre is incorporated to improve tensile strength and load-bearing capacity due to its fibrous nature. The combination of these materials results in a sustainable composite with improved mechanical performance.

3.2 Preparation of Materials

The preparation of materials is a crucial step to ensure good bonding and uniformity in the composite. Used tea powder is first collected and thoroughly washed to remove impurities such as dust and residual substances. It is then dried either under sunlight or in an oven to eliminate moisture content, as the presence of moisture can weaken the bonding between the matrix and reinforcement. After drying, the tea powder is ground into fine particles to achieve uniform size and better dispersion within the epoxy matrix. Ramie fibres are also cleaned to remove any dirt or unwanted materials and are cut into suitable lengths to ensure proper placement within the composite. Proper preparation of both tea powder and fibres helps in achieving improved interfacial bonding and overall mechanical performance.

3.3 Mould Preparation

A mould is prepared according to the required dimensions of the composite specimen. The mould material may be wood, plastic, or metal depending on availability. Before fabrication, the inner surface of the mould is cleaned and coated with a release agent such as oil, grease, or wax. This coating prevents the composite from sticking to the mould and ensures easy removal after curing. Proper mould preparation also helps in achieving a better surface finish and accurate dimensions of the final specimen.

3.4 Mixing of Materials

The mixing process plays a significant role in determining the quality of the composite. Epoxy resin and hardener are mixed in a specific ratio as recommended by the manufacturer to initiate the curing process. The mixture is stirred thoroughly to achieve a uniform consistency. After this, the prepared tea powder is gradually added to the epoxy mixture and mixed continuously to ensure even distribution of particles. Care is taken to avoid the formation of lumps and air bubbles during mixing, as these can create weak points in the composite. Proper mixing ensures uniform dispersion of reinforcement and enhances bonding between the matrix and filler.

3.5 Fabrication Process (Hand Lay-Up Technique)

The composite is fabricated using the hand lay-up technique, which is a simple and cost-effective method. Initially, a thin layer of the epoxy-tea powder mixture is poured into the prepared mould. Ramie fibres are then placed uniformly over this layer in the desired orientation. Additional layers of the mixture are applied to fully cover the fibres. This process can be repeated to form multiple layers if required. Gentle pressure is applied using a roller or flat tool to remove trapped air bubbles and ensure proper contact between the matrix and reinforcement. The hand lay-up method allows better control over material placement and is suitable for small-scale composite specimens.

3.6 Curing Process

After the fabrication is completed, the composite is left undisturbed for curing at room temperature. The curing process typically takes 24 to 48 hours, during which the epoxy resin reacts with the hardener to form a rigid, cross-linked polymer structure. Proper curing is essential to achieve maximum strength and durability. External disturbances, temperature variations, or insufficient curing time can affect the final properties of the composite. In some cases, post-curing at slightly elevated temperatures may be performed to enhance the mechanical properties further.

3.7 Specimen Finishing

Once the curing process is complete, the composite is carefully removed from the mould. The specimen is then trimmed to remove excess material and achieve the required dimensions. Surface finishing is carried out using tools such as files or sandpaper to smoothen the edges and improve appearance. Proper finishing ensures that the specimen is free from defects such as sharp edges or irregularities, making it suitable for further evaluation and testing. Accurate dimensions and good surface quality are important to ensure reliable results in subsequent analysis.

3.8 Flow Chart:



3.9 Cost Estimation:

S.NO	ITEM	QUANTITY	COST (INR)
1	Epoxy Resin LY556 (1 kg)	1	1000
2	Hardner HY951 (100g)	1	200
3	Ramine Fiber (500g)	1	300
4	Used tea powder (500g)	1	100
5	Total	4	1600

Table 3.5.1 Cost Estimation

The total estimated cost for the project is approximately ₹1600, which includes the cost of materials such as epoxy resin, hardener, used tea powder, and ramie fibre, along with fabrication and miscellaneous expenses. This cost-effective approach makes the project feasible for academic purposes while still providing reliable and meaningful results.

3.10 Experimental Samples Prepared for Tensile Testing:



Fig 3.6.1 Sample Photo

CHAPTER – 4
TESTING METHODS AND RESULT DISCUSSION

4.1 Testing Methods:

The testing of the fabricated waste particle reinforced epoxy composite specimens is carried out to evaluate their mechanical strength and behavior under load. In this project, tensile testing is selected as the primary method to determine the strength characteristics of the developed composite material. The prepared specimens made using epoxy resin, used tea powder, and ramie fibre are tested under identical conditions to ensure accurate and reliable comparison. The testing is conducted using a Universal Testing Machine (UTM), which applies a gradually increasing tensile load on the specimen until failure occurs. The maximum load sustained by the specimen is recorded and used to evaluate the tensile strength and overall performance of the composite. Proper care is taken to ensure uniform gripping and correct alignment of the specimen to avoid experimental errors.

In addition, proper calibration of the Universal Testing Machine (UTM) is ensured before conducting the experiment to obtain precise results. Each composite specimen is carefully placed between the grips of the machine to maintain proper alignment and to avoid uneven loading conditions. The loading rate is kept constant throughout the test to maintain consistency. During the test, parameters such as applied load and elongation are continuously monitored until fracture occurs. The mode of failure, such as brittle fracture, fibre pull-out, or matrix cracking, is also observed and recorded, as it provides valuable insights into the bonding and behavior of the composite material. This systematic testing procedure ensures that the obtained results are accurate and useful for analyzing the mechanical performance of the developed eco-friendly composite.

4.1.1 Tensile Test:

The tensile test is a fundamental mechanical test used to determine the strength properties of materials by applying an axial tensile force until the specimen fails. In this project, the developed waste particle reinforced epoxy composite specimen is subjected to tensile testing using a Universal Testing Machine (UTM). The specimen, having dimensions of 150 mm × 30 mm × 10 mm, is firmly fixed between the grips of the machine, ensuring proper alignment to avoid bending or uneven stress distribution. A gradually increasing tensile load is applied, and the corresponding deformation is observed until the specimen fractures.

The maximum load at which the composite specimen fails is recorded and used to evaluate its tensile strength. This test helps in understanding the ability of the developed composite material to withstand tensile forces and provides insight into its mechanical performance and durability. Since only one specimen is tested, the result gives a basic understanding of the strength characteristics of the material rather than a comparative analysis.

Further observation during testing includes the mode of failure, such as matrix cracking, fibre breakage, or fibre pull-out, which indicates the quality of bonding between epoxy, tea powder, and ramie fibre. Proper curing, uniform mixing, and good fibre distribution are expected to improve the strength and delay failure. Any defects such as voids or poor bonding may lead to early fracture. The results obtained from this tensile test help in evaluating the effectiveness of using waste tea powder and natural fibre in developing an eco-friendly composite material and provide a foundation for further improvement and analysis.



Fig 4.1.1 Universal Testing Machin

4.2 Test Results:

4.2.1 TEST SAMPLE NO. 2:

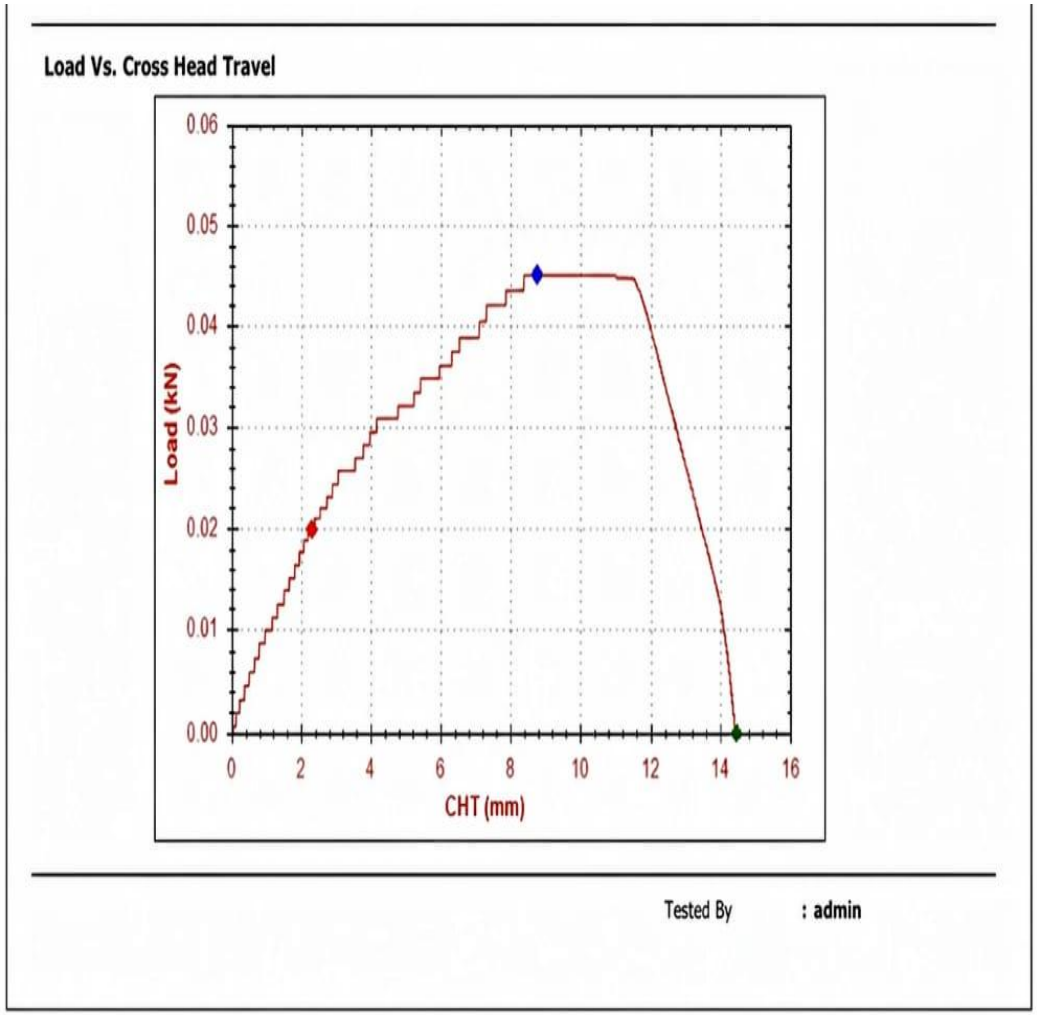


Fig 4.2.1 Graph of Sample

Output Data:

Load At Yield : 0.018 kN

Elongation At Yield : 2.312 mm

Yield Stress : 0.128 N/mm²

Load at Peak : 0.045 kN

Elongation at Peak : 8.265 mm

Tensile Strength : 0.321 N/mm²

Load At Break : 0.002 kN

Elongation At Break : 13.50mm

The tensile test conducted on the developed waste particle reinforced epoxy composite provides a detailed understanding of its mechanical behavior under axial loading. The composite specimen, made using epoxy resin, used tea powder, and ramie fibre, was tested using a Universal Testing Machine (UTM) to evaluate its strength and deformation characteristics. From the obtained results, the specimen reaches a yield load of 0.018 kN at an elongation of 2.312 mm, corresponding to a yield stress of 0.128 N/mm². This point indicates the transition from elastic to plastic deformation, where the material starts to undergo permanent deformation. The relatively low yield stress suggests that the composite begins to deform at lower loads, which may be influenced by the presence of particulate filler and natural fibres affecting the bonding efficiency.

As the applied load increases, the composite continues to deform steadily, showing a gradual rise in the load-carrying capacity. The specimen reaches its maximum (peak) load of 0.045 kN at an elongation of 8.265 mm, which represents the highest load the material can withstand before the initiation of failure. The corresponding tensile strength

is calculated as 0.321 N/mm², indicating moderate strength characteristics of the developed composite. This level of strength is suitable for lightweight and low-load applications, but it is lower compared to conventional synthetic composites, mainly due to the use of natural and waste-based reinforcement materials.

After reaching the peak load, the material shows a noticeable drop in load-carrying capacity, indicating the initiation and propagation of cracks within the composite structure. The specimen ultimately fails at a break load of 0.002 kN with an elongation of 13.500 mm. The relatively higher elongation at break compared to the yield elongation suggests that the composite exhibits some degree of ductile behavior before final fracture. This behavior can be attributed to the presence of ramie fibres, which help in resisting crack propagation and provide limited energy absorption capacity. However, the sudden reduction in load after the peak indicates that the material does not sustain significant plastic deformation and eventually undergoes failure due to matrix cracking, fibre breakage, or fibre pull-out.

The load vs. cross-head travel graph shows a smooth and gradual increase in load initially, which indicates effective stress transfer between the epoxy matrix, tea powder particles, and ramie fibres. This suggests that the mixing and fabrication process achieved a reasonably uniform distribution of materials. However, minor fluctuations in the curve may indicate the presence of small defects such as voids or weak bonding regions within the composite. The sharp drop in the curve after the peak load clearly represents the failure point of the material, highlighting the brittle nature of the epoxy matrix combined with partial ductility introduced by the fibres.

The overall performance of the composite is influenced by several factors, including the quality of mixing, uniform distribution of tea powder and fibres, proper curing, and interfacial bonding between the matrix and reinforcement. Tea powder acts mainly as a filler material, contributing to stiffness but potentially reducing bonding strength if present in excess. On the other hand, ramie fibre plays a significant role in enhancing tensile strength and improving deformation resistance by bridging cracks and delaying

failure.

In conclusion, the tensile test results demonstrate that the developed waste particle reinforced epoxy composite exhibits moderate strength with noticeable deformation before failure. While the material may not be suitable for high-load structural applications, it shows good potential for lightweight, low-cost, and eco-friendly engineering applications. The results also highlight the importance of optimizing material composition and fabrication techniques to further improve strength, bonding, and overall mechanical performance of the composite.

CHAPTER – 5
CONCLUSION

The present study focused on the development and strength evaluation of a waste particle reinforced epoxy composite using used tea powder and ramie fibre as reinforcement materials. The composite was fabricated using the hand lay-up technique under controlled conditions to ensure uniform mixing and proper bonding between the matrix and reinforcement. Epoxy resin was selected as the base material due to its strong adhesive properties and good mechanical performance, while tea powder and ramie fibre were incorporated to enhance strength and promote sustainability. The methodology provided a systematic approach to understand how material composition, mixing quality, and curing conditions influence the overall performance of the composite.

The tensile test results indicated that the developed composite exhibits moderate strength and good stiffness suitable for low to moderate load applications. The presence of ramie fibre contributed to improved tensile strength and load-bearing capacity, while tea powder acted as a filler that enhanced stiffness and reduced material cost. It was also observed that proper mixing and curing play a crucial role in achieving better mechanical properties, whereas poor bonding, moisture content, or uneven distribution can reduce strength and lead to failure. Overall, the study demonstrates that waste tea powder and natural fibres can be effectively utilized to develop eco-friendly and cost-effective composite materials. The findings highlight the potential of such composites for lightweight engineering applications and suggest that further improvements can be achieved by optimizing material composition and fabrication techniques.

CHAPTER – 6
FUTURE SCOPE

- Use of advanced reinforcement materials such as hybrid natural fibres or synthetic fibres (glass/carbon) along with ramie fibre to further improve strength and durability of the composite.
- Optimization of material composition by varying the percentage of epoxy resin, tea powder, and fibre content to achieve maximum tensile strength and better performance.
- Improvement in fabrication techniques by exploring methods such as compression moulding or vacuum-assisted processes to reduce voids and enhance bonding.
- Application of statistical tools and Design of Experiments (DOE) to systematically analyze the effect of different parameters on mechanical properties.
- Conducting additional mechanical tests such as compression, flexural, impact, and hardness tests for a more comprehensive evaluation of the composite material.
- Surface treatment of natural fibres (chemical treatment) to improve fibre-matrix adhesion and reduce moisture absorption.
- Comparison of the developed composite with conventional materials such as plastics, wood, or synthetic fibre composites to evaluate performance advantages.
- Development of hybrid composites by combining multiple waste materials or different natural fibres to enhance overall properties.
- Study of long-term performance, including durability, moisture resistance, and environmental effects on the composite.
- Exploration of real-time engineering applications such as panels, boards, and lightweight structural components using the developed eco-friendly composite material.

The scope of this project lies in developing and analyzing the mechanical performance of a waste particle reinforced epoxy composite using used tea powder and

ramie fibre. By fabricating composite specimens through the hand lay-up technique and controlling factors such as material composition, mixing quality, and curing conditions, the study provides a clear understanding of how these parameters influence tensile strength and deformation behavior of the developed material.

This work is not limited to a single specimen but can be extended to various eco-friendly composite applications in light engineering fields such as panels, boards, and structural components. The results help in identifying suitable material combinations and fabrication practices, which can be applied in industries where low-cost, lightweight, and sustainable materials are required.

Furthermore, this project opens opportunities for future advancements such as the use of hybrid reinforcements, improved fabrication techniques, and optimized material proportions to enhance strength and durability. The study also provides a foundation for comparing this composite with conventional materials like plastics, wood, and synthetic fibre composites. With the increasing demand for sustainable materials, the findings of this project contribute to the development of environmentally friendly and cost-effective composite products.

In addition, the scope of this project extends to educational and research applications, where it serves as a practical demonstration of how material composition and fabrication methods affect the performance of composite materials. The experimental approach used in this study can act as a reference for future projects involving natural fibre composites and waste material utilization. It also supports the development of optimized material design and manufacturing strategies for producing lightweight and efficient components.

By integrating material selection, fabrication, and evaluation, this project contributes to a better understanding of composite material technology and its potential in modern engineering applications.

CHAPTER – 7
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